Room Booking API: Requirements

Nov 2, 2026 · @MannChealy

1. Purpose and scope

This version adds room booking to the existing NBC Meeting Room API: staff book rooms, Pitika and room owners approve, IT prepares the room, and the booker gets a door pass. It is built on the rooms, places, organisation and users that already exist.

The rules come from three sources: the booking ERD (Booking (3).drawio), the clickable mockup (frontend_mockup) and the old full spec (Testing_Claude/docs/REQUIREMENTS.md). The decisions were reviewed with the boss on 2 Nov 2026 (doc "Room Booking: Decisions for Review"). Where this spec and the old spec differ, this spec wins.

In this version

- Bookings with 1 to 20 sessions, and a clash check on every session.
- Four booking routes, decided by the system when a booking is sent (section 4).
- Pitika review, room owner review with shorter times, and expiry.
- Food (catering) and IT support requests, approved with the booking.
- IT tickets: the IT Lead assigns IT Staff and presses Ready.
- Door pass, calendar file (.ics), cancel, small and big edits.
- Admin lists for public holidays, catering packages and IT services.

Not in this version

- Screens. The booker, Pitika, owner and IT screens get their own spec later.
- Notifications: no bell, no email, no Telegram. People see changes when they open a booking.
- Prices and billing. Rooms, food and IT are free.
- Office closures and room repair dates. Only public holidays, and they only warn.
- A backup approver for room owners.
- Reports, charts and an audit log.

2. Glossary

Word	Meaning
Booking	One request to use one room. It has a title, attendees and 1 to 20 sessions.
Session	One date with a start and end time inside a booking.
Booker	The user who sends the booking. Also called the requester.
Route	The approval path the system picks when the booking is sent: R1 to R4 (section 4).
Shared room	A room that is not private.
Private room	A room marked private. It always has a room owner.
Room owner	The user set as owner of a room. Reviews bookings in their private rooms.
Pitika	A user with the Pitika role. Reviews bookings for rooms in the province of their division.
Extras	Food (catering) and IT support asked for in a booking.
IT ticket	The IT work for one booking. The IT Lead assigns IT Staff to it.
Door pass	A code ("NBC-" and 5 digits) to open the room, shown once the booking is confirmed.
Reference code	A short code ("NBC-" and 5 digits) to name a booking on the phone or on paper. It is not the door pass.
Blocking	A session that holds its time in the room. Every session blocks unless its booking is rejected or cancelled.

3. Roles and permissions

The six roles already exist. A user can hold several. Every check runs on the server; hiding a button is not enough.

Role	In booking
Staff	Books rooms; sees, edits and cancels their own bookings.
Room owner	Reviews bookings in the private rooms they own. Books their own room with no review.
Pitika	Reviews bookings for rooms in their province. Sees the food orders there. Can cancel with a reason.
IT Lead	Sees every IT ticket, assigns IT Staff and presses Ready.
IT Staff	Sees the IT tickets they are assigned to.
Admin	Sees every booking, can cancel with a reason, and keeps the holidays, catering packages and IT services lists.

"Their province" for a Pitika = the province of the office of their division. "Own rooms" for a room owner = rooms where they are the owner.

Action	Staff	Room owner	Pitika	IT Lead	IT Staff	Admin
See rooms and their schedule	Yes	Yes	Yes	Yes	Yes	Yes
Send a booking; see, edit and cancel own bookings	Yes	Yes	Yes	Yes	Yes	Yes
See any booking's details	-	Own rooms	Their province	Bookings with IT	Assigned tickets	All
See the title of another person's private-room booking	-	Own rooms	Their province	Bookings with IT	Assigned tickets	All
Approve or reject step 1	-	-	Their province	-	-	-
Approve or reject step 2, shorten times	-	Own private rooms	-	-	-	-
Assign IT Staff, press Ready	-	-	-	Yes	-	-
See the door pass	Own, when confirmed	Own rooms	Their province	-	-	All

Action	Staff	Room owner	Pitika	IT Lead	IT Staff	Admin
Cancel anyone's booking, with a reason	-	-	Their province	-	-	All
See food orders	-	-	Their province	-	-	All
Keep holidays, catering packages, IT services	-	-	-	-	-	Yes

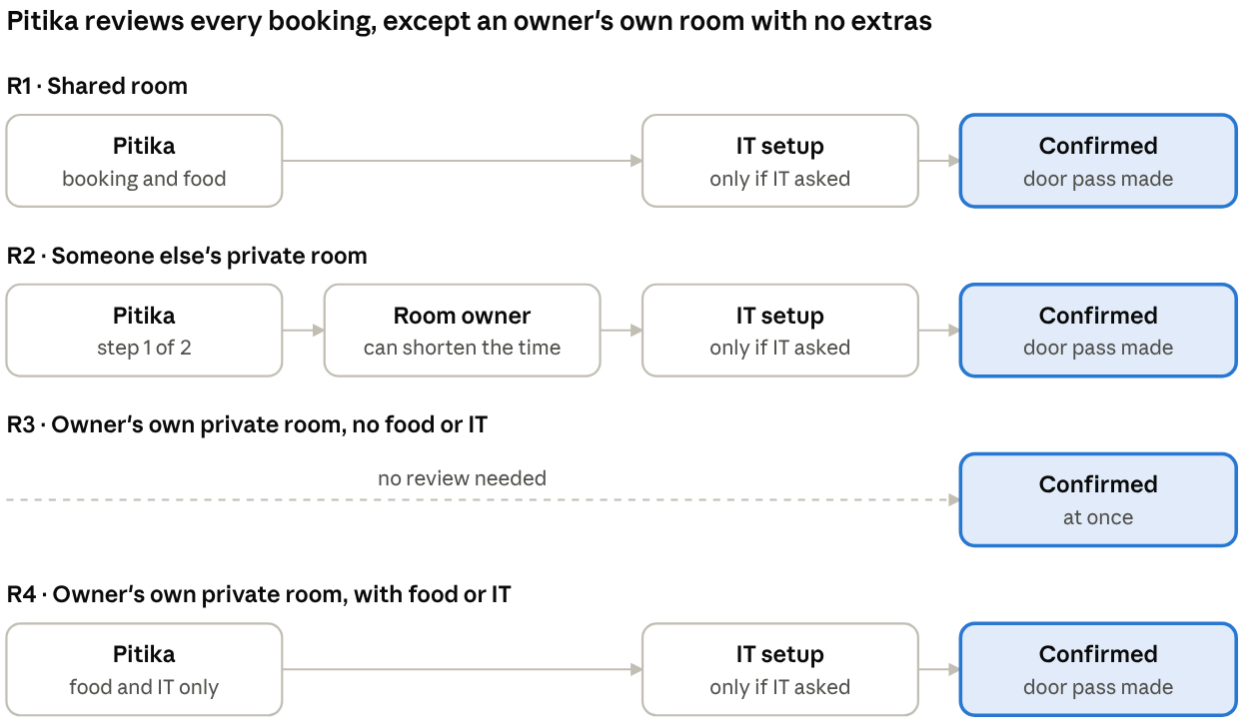
Who may send a booking is an open question (section 12): this table assumes every active user.

4. Booking routes and flows

The system picks the route when the booking is sent, from two facts: the room and the booker, and whether food or IT was asked for. The booker never chooses it. A big edit picks the route again.

Route	Booking	Extras	First status	Reviews
R1	Shared room	Any	PENDING_PITIKA	Pitika
R2	Someone else's private room	Any	PENDING_PITIKA	Pitika, then room owner
R3	Owner's own private room	None	CONFIRMED	None
R4	Owner's own private room	Food and/or IT	PENDING_PITIKA	Pitika checks food and IT only

After the last review: if IT was asked, the booking becomes SETUP_IN_PROGRESS and the IT ticket goes to the IT Lead. Otherwise it becomes CONFIRMED. Food never holds a booking in setup.



booking routes · R1 to R4

At each review step the reviewer can reject, always with a reason. A booking still waiting when its first session starts is rejected by the system. The booker can cancel until the first session starts.

5. Statuses and transitions

A booking is always in one of six statuses. Only the moves in the transition table are allowed; any other move is refused with `409 INVALID_TRANSITION`.

Code	Shown as	Holds its time?	Final?
PENDING_PITIKA	Waiting for Pitika	Yes	No
PENDING_OWNER	Waiting for the room owner	Yes	No
SETUP_IN_PROGRESS	Setting up	Yes	No
CONFIRMED	Confirmed	Yes	No
REJECTED	Rejected	No	Yes
CANCELLED	Cancelled	No	Yes

After the door pass ends (end of the last session), a confirmed booking is shown as **Completed**. This is worked out when read, never saved, and a completed booking can no longer be edited or cancelled.

#	From	Event	Who	Must be true	To	Also happens
T1	(new)	Send, route R1, R2 or R4	Booker	Rules in 7.1 to 7.3 pass	PENDING_PITIKA	IT ticket made (not yet forwarded)
T2	(new)	Send, route R3	Booker = owner	Rules pass, no extras	CONFIRMED	Door pass made; Pitika review saved as NOT_REQUIRED
T3	PENDING_PITIKA	Pitika approves	Pitika	Route R2; first session not started	PENDING_OWNER	Food approved
T4	PENDING_PITIKA	Pitika approves	Pitika	Route R1 or R4; IT asked; first session not started	SETUP_IN_PROGRESS	Food approved; IT ticket forwarded
T5	PENDING_PITIKA	Pitika approves	Pitika	Route R1 or R4; no IT; first session not started	CONFIRMED	Food approved; door pass made
T6	PENDING_PITIKA	Pitika rejects	Pitika	Reason given	REJECTED	Sessions stop blocking
T7	PENDING_OWNER	Owner approves	Room owner	Shorter times valid (7.5); IT asked	SETUP_IN_PROGRESS	IT ticket forwarded
T8	PENDING_OWNER	Owner approves	Room owner	Shorter times valid; no IT	CONFIRMED	Door pass made
T9	PENDING_OWNER	Owner rejects	Room owner	Reason given	REJECTED	Sessions stop blocking

#	From	Event	Who	Must be true	To	Also happens
T10	SETUP_IN_PROGRESS	Assign staff	IT Lead	At least 1 active IT Staff	(same)	Ticket status ASSIGNED
T11	SETUP_IN_PROGRESS	Press Ready	IT Lead	Staff assigned	CONFIRMED	Ticket status READY; door pass made
T12	Any not final	Cancel	Booker	First session not started	CANCELLED	Sessions stop blocking; IT ticket and food dropped
T13	Any not final	Small edit	Booker	First session not started	(same)	Only the small fields change
T14	Any not final	Big edit	Booker	First session not started; new sessions pass 7.1	Route again (T1 or T2)	Reviews cleared; IT ticket un-forwarded and staff removed; door pass removed
T15	PENDING_PITIKA or PENDING_OWNER	Expire	System	First session has started	REJECTED	Rejected by SYSTEM: "Expired: not approved before the meeting time."
T16	Any not final	Cancel by staff	Pitika or Admin	Last session not ended; reason given	CANCELLED	As T12; finished sessions stay in history

Two people at once. Every move checks the `version` number in the same database transaction: `UPDATE bookings SET status = …, version = version + 1 WHERE id = … AND version = …`. If no row changes, the answer is `409 STALE_VERSION` ("This booking changed. Reload it."). Two reviewers can never both succeed.

6. Data model

The tables follow the booking ERD. `users` and `rooms` already exist. Rows marked **(added)** are not in the ERD yet; section 12 asks to confirm them.

6.1 Enums

Enum	Values
BookingStatus	PENDING_PITIKA, PENDING_OWNER, SETUP_IN_PROGRESS, CONFIRMED, REJECTED, CANCELLED
MeetingType	INTERNAL, EXTERNAL, INTERNAL_WITH_GOVERNOR, EXTERNAL_WITH_GOVERNOR
ReviewStage	PITIKA, OWNER
ReviewDecision	APPROVED, REJECTED, NOT_REQUIRED
RejectedBy	PITIKA, OWNER, SYSTEM
ITStatus	RECEIVED, ASSIGNED, READY

6.2 bookings

Field	Type	Notes
id	int	Auto increment
reference_code	text(20), unique	"NBC-" and 5 random digits, retried on a clash
requester_id	FK users	The booker
room_id	FK rooms	
requester_info	JSON	Copy at send time: name_en, name_kh, staff_id, position, division, department, office, email, phone
meeting_title	text(200)	Required, 3 to 200 letters
meeting_type	MeetingType	Required
meeting_purpose	text(1000), optional	Agenda or notes
attendees	int	1 to 100
private_justification	text(1000), optional	Required for route R2 only
route	R1 to R4 **(added)**	Set at send and at every big edit
status	BookingStatus	

Field	Type	Notes
cancelled_at	datetime, optional	
cancelled_by_id	FK users, optional **(added)**	Booker, Pitika or Admin
cancel_reason	text(500), optional	Required when Pitika or Admin cancels
version	int, default 1	Raised on every change (section 5)
created_at, updated_at	datetime	

6.3 booking_sessions

Field	Type	Notes
id	int	
booking_id	FK bookings	1 to 20 per booking
date	date	
start_time, end_time	time	Inside the room's opening hours
requested_start_time, requested_end_time	time, optional **(added)**	The booker's times, kept when the owner shortens a session
is_blocking	bool	True unless the booking is rejected or cancelled. Index (room, date, is_blocking) for clash checks.

The booking's shown date and time = its first session (by date, then start time).

6.4 Review and access

Table	Fields	Notes
booking_reviews	booking_id, stage, decision, reviewer_id, notes(1000), reviewed_at	One row per review. A big edit deletes the old rows. R3 saves one PITIKA row as NOT_REQUIRED.
booking_rejections	booking_id (unique), rejected_by, actor_user_id, reason(1000), decided_at	One per rejected booking. actor_user_id is empty for SYSTEM, so it must allow NULL.

Table	Fields	Notes
booking_door_passes	booking_id (unique), passcode, expires_at	Made when the booking becomes CONFIRMED. Replaced on a big edit after confirmation.
booking_participants	booking_id, email	Lowercase, unique per booking, at most 50.

6.5 Services

Table	Fields	Notes
catering_orders	booking_id (unique), packages JSON, remarks(1000), approver_notes(1000), decided_by, decided_at	packages = copy of [{id, name}] at send time. Servings = attendees; delivery = first session start. decided_by = the Pitika who approved the booking.
it_tickets	booking_id (unique), requested_items JSON, special_requirements(1000), status, technician_notes(1000), forwarded_at, assigned_at, ready_at	requested_items = copy of [{id, name}]. forwarded_at is set at final approval; IT sees only forwarded tickets.
it_ticket_staff	ticket_id, user_id	Unique pair. The user must be an active IT Staff.

6.6 Catalogs

Table	Fields	Notes
catering_packages	name(200), description(1000), sort_order, is_active, version, timestamps	Admin list. Inactive ones cannot be picked for new bookings.
it_services	name(200), description(1000), sort_order, is_active, version, timestamps	Admin list, same rules.
public_holidays	holiday_date (unique), name_km(255), name_en(255), remarks, timestamps	Admin list. Only gives a warning (7.10).

7. Business rules

Each rule has an ID so code, tests and questions can point at it. The numbers (15 minutes, 20 sessions, 90 days, 100 attendees) are API settings with these defaults, not fixed in code.

7.1 Scheduling

ID	Rule
BR-SCH-01	Every session fits inside the room's own opening and closing time. Message: "This room is open from 08:00 to 17:00."
BR-SCH-02	End is after start, and a session is at least 15 minutes.
BR-SCH-03	No session in the past. Today: the start is at or after the current time (Phnom Penh time).
BR-SCH-04	A session clashes with another in the same room and date when new start < other end AND new end > other start. Back-to-back sessions (one ends 10:00, next starts 10:00) are allowed.
BR-SCH-05	Only blocking sessions count: every booking that is not rejected or cancelled, including waiting ones.

ID	Rule
BR-SCH-06	Sessions inside one booking must not overlap each other.
BR-SCH-07	All sessions are checked together. If one fails, nothing is saved, and the error names that session.
BR-SCH-08	Clash, past-time and holiday checks run on the server when the booking is saved, in one transaction that locks the room row (`SELECT … FOR UPDATE`).
BR-SCH-09	Only bookable rooms: the room, its floor, building and office are all active.
BR-SCH-10	At most 20 sessions per booking, and no session more than 90 days ahead.
BR-SCH-11	Weekends are allowed.
BR-SCH-12	A session on a public holiday is allowed, with a warning (7.10).
BR-SCH-13	Routes R1, R2 and R4 are refused when the room's province has no active Pitika: "No reviewer for this office yet." R3 needs no Pitika and is allowed.

7.2 Privacy

ID	Rule
BR-PRV-01	In a room's schedule, a booking in a private room is shown as "Reserved (Private Session)" to users who may not see it (section 3). Title, booker and purpose are hidden.
BR-PRV-02	Clash messages follow the same rule: "This room is already booked from 08:30 to 10:00 for Reserved (Private Session)."
BR-PRV-03	The door pass is never sent before the booking is confirmed, and only to the users in section 3.

7.3 Booking form

ID	Rule
BR-FRM-01	At least one session.
BR-FRM-02	Meeting title (3 to 200 letters) and meeting type are required.
BR-FRM-03	Attendees: a whole number from 1 to 100, and at least the number of participants plus 1 (the booker).
BR-FRM-04	More attendees than room seats is allowed, with a warning: "This room has 12 seats. You entered 20 attendees."
BR-FRM-05	Participant emails: valid, saved lowercase, no repeats, at most 50.
BR-FRM-06	requester_info is copied from the booker's profile when the booking is sent, so later profile changes do not change old bookings.
BR-FRM-07	Route R2 (another person's private room) needs a private justification.
BR-FRM-08	Food: at least one active catering package, plus optional remarks (for example "2 vegetarian, no pork"). IT: at least one active IT service, plus optional special requirements.

7.4 Pitika review

ID	Rule
BR-APR-01	A Pitika acts only on PENDING_PITIKA bookings for rooms in their province. Several Pitika share that queue; the first to act decides.
BR-APR-02	Rejecting always needs a reason.
BR-APR-03	A booking whose first session has started cannot be approved; it can still be rejected. Expiry (T15) rejects it anyway.

ID	Rule
BR-APR-04	A Pitika can never skip the room owner on route R2: approving moves it to PENDING_OWNER.
BR-APR-05	On route R4 (owner's own room with extras), the Pitika checks only the food and IT.
BR-APR-06	One Pitika approval also approves the food: catering_orders.decided_by and decided_at are set at the same moment.
BR-APR-07	Optional approval notes are kept and shown to the next step (owner, IT Lead) and as kitchen notes on the food order.

7.5 Room owner review

ID	Rule
BR-OWN-01	The owner acts only on PENDING_OWNER bookings in their own private rooms.
BR-OWN-02	The owner may make a session shorter. The new time sits inside the requested time, is at least 15 minutes, and does not clash. The requested time is kept. The freed time can be booked at once.
BR-OWN-03	With many sessions, each can be shortened on its own, but the owner approves or rejects the whole booking.
BR-OWN-04	There is no backup approver. If the owner does not act, the booking expires (T15).
BR-OWN-05	Waiting bookings follow the room: when the Admin sets a new owner, the new owner sees them at once.

7.6 IT support

ID	Rule
BR-IT-01	The IT ticket is made when the booking is sent, but IT sees it only after the final approval (forwarded_at is set) and while the booking is not rejected or cancelled.
BR-IT-02	The IT Lead assigns one or more active IT Staff and can change them later.
BR-IT-03	Only the IT Lead presses Ready, and only after staff are assigned. This confirms the booking (T11).
BR-IT-04	IT Staff see the tickets they are assigned to; they cannot change them.
BR-IT-05	Setup is planned 30 minutes before the first session (shown as information).

7.7 Door pass

ID	Rule
BR-DP-01	Format "NBC-" and 5 digits (10000 to 99999), from a secure random source, unique among active door passes, and never equal to the reference code.
BR-DP-02	Made when the booking becomes CONFIRMED. It ends at the end of the last session.
BR-DP-03	After it ends, the API still returns it with "expired": true.
BR-DP-04	A big edit removes it; a new one is made when the booking is confirmed again.
BR-DP-05	It is not linked to a real door lock, and it is not emailed in this version.

7.8 Cancel

ID	Rule
BR-CAN-01	The booker can cancel any booking that is not final, until the first session starts.
BR-CAN-02	A Pitika (their province) or the Admin can cancel until the last session ends, always with a reason.
BR-CAN-03	Cancelling frees the time at once and drops the food order and IT ticket.

7.9 Edit

ID	Rule
BR-EDT-01	Only the booker edits, only while the booking is not final, and only before the first session starts.
BR-EDT-02	**Small fields** keep the approval: title, meeting type, purpose, attendees, participants, food remarks, IT special requirements.
BR-EDT-03	**Big fields** start the approval again (T14): room, any session, food on or off or its packages, IT on or off or its services.
BR-EDT-04	A big edit is all or nothing. New sessions are checked against every other booking (not its own old sessions). If one fails, the booking stays exactly as it was.
BR-EDT-05	The answer says which kind it was: "edit_type": "MINOR" or "MAJOR".

7.10 Public holidays

ID	Rule
BR-HOL-01	The Admin keeps the list: one row per date, Khmer name required, English name optional.

ID	Rule
BR-HOL-02	Booking on a holiday is allowed. The check and the save answers carry a warning: {"code": "PUBLIC_HOLIDAY", "date": "2026-11-09", "name": "Independence Day"}.
BR-HOL-03	Adding a holiday changes no existing booking.

7.11 Field limits

Field	Limit
Meeting title	3 to 200 letters
Purpose, justification, notes, remarks, requirements	At most 1000 letters
Reject reason	5 to 1000 letters
Cancel reason	5 to 500 letters
Participant emails	At most 50 per booking
Catering package and IT service names	2 to 200 letters

Text is trimmed before checks. HTML is never accepted; text is saved as plain text.

8. API endpoints

Same style as the current API: base path `/api/v1`, JSON in snake_case, sign-in with the existing access token, lists with `page` (default 1) and `size` (default 20, at most 100) that answer `{"items": […], "page": 1, "size": 20, "total": 240}`. Every request that changes a booking sends its `version`.

8.1 Booker

Method	Path	What it does
GET	/rooms/{id}/schedule?from=&to=	Blocking sessions of one room between two dates, private ones masked (BR-PRV-01). At most 31 days.

Method	Path	What it does
POST	/bookings/check	Body: room_id, sessions, has_catering, has_it. Answers the route (R1 to R4), the steps the booking will go through, the first problem found, and any warnings. Saves nothing.
POST	/bookings	Sends a booking (T1 or T2). Answers 201 with the full booking, its route, status and warnings.
GET	/bookings/mine?status=&q=	The booker's own bookings, newest first.
GET	/bookings/{id}	One booking, for the users allowed in section 3.
PATCH	/bookings/{id}	Edit (T13 or T14). Same body as POST plus version. Answers edit_type MINOR or MAJOR and the new status.
POST	/bookings/{id}/cancel	Body: reason (required for Pitika and Admin), version. T12 or T16.
GET	/bookings/{id}/door-pass	The door pass, its end time and "expired". Only when CONFIRMED.
GET	/bookings/{id}/ics	A calendar file with every session.

POST /bookings body

```
{
  "room_id": 4,
  "sessions": [
    {"date": "2026-11-10", "start_time": "09:00", "end_time": "10:30"},
    {"date": "2026-11-11", "start_time": "14:00", "end_time": "15:00"}
  ],
  "meeting_title": "Q4 budget review",
  "meeting_type": "INTERNAL",
  "meeting_purpose": "Check the budget plan for 2027",
  "attendees": 12,
  "participant_emails": ["sok.dara@nbc.gov.kh"],
  "private_justification": null,
  "catering": {"package_ids": [1, 2], "remarks": "2 vegetarian"},
  "it": {"service_ids": [1], "special_requirements": "Need a Mac HDMI
adapter"}
}
```

`catering` and `it` are `null` when not asked for. The server picks the route; the body never sends it.

8.2 Pitika

Method	Path	What it does
GET	/pitika/bookings?status=&from=&to=&room_id=&q=	Bookings for rooms in the Pitika's province. from and to match any session date.
POST	/pitika/bookings/{id}/approve	Body: notes, version. The server picks T3, T4 or T5 from the route.
POST	/pitika/bookings/{id}/reject	Body: reason, version. T6.
GET	/pitika/catering?date=	Food orders of approved bookings on one day: packages, servings, room, delivery time, remarks.

8.3 Room owner

Method	Path	What it does
GET	/owner/bookings?status=&room_id=&q=	Bookings in the owner's private rooms.
POST	/owner/bookings/{id}/approve	Body: sessions (optional list of {session_id, start_time, end_time} to shorten), notes, version. T7 or T8.
POST	/owner/bookings/{id}/reject	Body: reason, version. T9.

8.4 IT

Method	Path	What it does
GET	/it/tickets?status=&q=	IT Lead: every forwarded ticket. IT Staff: only tickets they are assigned to.
PUT	/it/tickets/{id}/staff	IT Lead. Body: user_ids, technician_notes, version. T10.
POST	/it/tickets/{id}/ready	IT Lead. Body: version. T11.

8.5 Catalogs and holidays

Every signed-in user can read these lists (the booking form needs them). Only the Admin changes them.

Method	Path	What it does
GET, POST	/catering-packages	List; add one
PATCH	/catering-packages/{id}	Edit, make active or inactive
GET, POST	/it-services	List; add one
PATCH	/it-services/{id}	Edit, make active or inactive
GET, POST	/public-holidays?year=	List; add one
PATCH, DELETE	/public-holidays/{id}	Edit; delete (no booking points at a holiday)

8.6 Expiry job

A job runs every minute. It rejects (T15) every PENDING_PITIKA or PENDING_OWNER booking whose first session has started. It runs in one process only.

9. Errors

Errors keep the current shape: `{"error": {"code": "SCHEDULE_CONFLICT", "message": "…", "fields": {…}}}`. The existing codes stay (VALIDATION_ERROR 422, FORBIDDEN 403, NOT_FOUND 404, INACTIVE_REFERENCE and others). Booking adds these:

Code	HTTP	When	Example message
SCHEDULE_CONFLICT	409	A session clashes with another booking	"This room is already booked from 08:30 to 10:00 for Reserved (Private Session)."
TIME_PASSED	409	A session is in the past, or a review comes after the first session started	"This session starts at 09:00, which has passed."
OUTSIDE_ROOM_HOURS	422	A session is outside the room's opening hours	"This room is open from 08:00 to 17:00."

Code	HTTP	When	Example message
ROOM_NOT_BOOKABLE	409	The room, its floor, building or office is inactive	"This room cannot be booked right now."
NO_REVIEWER	409	The room's province has no active Pitika (BR-SCH-13)	"No reviewer for this office yet."
INVALID_TRANSITION	409	A move not in the transition table	"This booking is already confirmed."
STALE_VERSION	409	Someone changed the booking first	"This booking changed. Reload it."
OWNER_REQUIRED	422	A private room is saved with no owner (section 10)	"A private room needs an owner."
OWNER_HAS_PRIVATE_ROOMS	409	Making a room owner inactive, or removing the role, while they own a private room	"Assign a new owner for 2 private rooms first."

Limits on sessions, attendees, emails and text lengths use VALIDATION_ERROR with `fields`, for example `{"sessions.2.end_time": "A session is at least 15 minutes."}`.

Warnings are not errors. The check and save answers carry a list such as `"warnings": [{"code": "PUBLIC_HOLIDAY", …}, {"code": "OVER_CAPACITY", "seats": 12, "attendees": 20}]`.

10. Changes to the current API

These rules change things that are already built and used by the admin dashboard. The dashboard must show the new messages.

Where	Today	After
Create or edit a room	A private room can have no owner.	Refused with OWNER_REQUIRED.

Where	Today	After
Make a user inactive, or remove their Room owner role	Their rooms lose their owner.	Refused with OWNER_HAS_PRIVATE_ROOMS while they own a private room. Shared rooms still lose their owner, as today.
Change a room's owner	Only the room changes.	Waiting bookings follow the room: the new owner sees them (BR-OWN-05).
Change a private room to shared	Allowed.	Refused while the room has PENDING_OWNER bookings: "Resolve 3 bookings waiting for the room owner first."
Lower a room's seats	Allowed.	Still allowed; existing bookings are not changed.
Change a room's opening hours	Allowed.	Still allowed; existing bookings are not checked again.
Make a room, floor, building or office inactive	Allowed; it stops being bookable.	Still allowed; open question 3 asks what happens to its future bookings.

11. Decisions log

Every place where the sources disagreed or said nothing has one decision here. "Boss review" = decided in the review of 2 Nov 2026. "Writer's call" = chosen while writing this spec; change it with a comment.

ID	Topic	Decision	Source	Where
D-01	Catering approval	One Pitika approval approves the booking and its food	Boss review	BR-APR-06
D-02	IT roles	IT Lead assigns IT Staff and presses Ready	Boss review (changed)	BR-IT-02, BR-IT-03
D-03	Booking hours	Each room's own opening hours	Boss review	BR-SCH-01
D-04	Holidays	Public holidays only; allowed with a warning	Boss review	7.10

ID	Topic	Decision	Source	Where
D-05	Owner step	A private room must have an owner, who reviews after Pitika	Boss review	Route R2, section 10
D-06	Backup approver	None; the booking expires	Boss review	BR-OWN-04
D-07	Pitika queue	One queue per province	Boss review (changed)	BR-APR-01
D-08	No Pitika in a province	New bookings needing Pitika are refused	Boss review	BR-SCH-13
D-09	Meeting type	4 values, label only	Boss review (note)	6.1
D-10	Notifications	None in this version	Boss review	Section 1
D-11	Scope	Booking API only	Boss review	Section 1
D-12	Owner leaves	Refused while owning a private room	Boss review	Section 10
D-13	Routes, statuses, transitions	As the old spec, "Manager" renamed Pitika	Old spec	Sections 4, 5
D-14	Time limits	15 minutes, 20 sessions, 90 days, weekends allowed	Old spec	7.1
D-15	Edit and cancel	Small and big edits; cancel cut-offs	Old spec	7.8, 7.9
D-16	Expiry	System rejects at first session start	Old spec	T15
D-17	Door pass	Made at confirm, not emailed, not linked to a lock	Old spec, changed by D-10	7.7
D-18	Pending bookings block	Waiting bookings hold their time	Old spec and mockup	BR-SCH-05
D-19	Food after approval	Food never holds a booking in setup; only IT does	Old spec	T4, T5

ID	Topic	Decision	Source	Where
D-20	Review history	A big edit deletes old review rows (no audit log yet)	Writer's call	6.4
D-21	Private justification	Required for route R2 only	Writer's call	BR-FRM-07
D-22	Food list	Pitika sees the day's food orders for their province	Writer's call	8.2
D-23	Session grid	The API takes any minute; a 30-minute grid is for screens	Writer's call	7.1

12. Tests and open questions

12.1 Tests that must pass

Each item is at least one API test (pytest).

- [ ] Each route R1 to R4 from send to CONFIRMED, with and without IT.
- [ ] Each transition T1 to T16, and one move that is not allowed (INVALID_TRANSITION).
- [ ] Clash: overlap refused, back-to-back allowed, a rejected or cancelled booking does not block, a waiting one does.
- [ ] Two sessions in one booking that overlap are refused; one bad session saves nothing.
- [ ] Room hours, 15-minute minimum, past time, 20 sessions, 90 days.
- [ ] Two Pitika approve the same booking at once: one wins, one gets STALE_VERSION.
- [ ] A Pitika cannot see or act on a room in another province; NO_REVIEWER when a province has no Pitika; R3 still works.
- [ ] Owner shortens one of two sessions; the freed time can be booked; a longer or clashing time is refused.
- [ ] Expiry job rejects a waiting booking after its first session starts.
- [ ] Door pass: hidden before confirm, shown after, "expired" after the last session, replaced after a big edit.
- [ ] Private booking shown as "Reserved (Private Session)" to others, in the schedule and in clash messages.
- [ ] Small edit keeps the status; big edit routes again; a failed big edit changes nothing.
- [ ] Cancel cut-offs for the booker, Pitika and Admin.
- [ ] Holiday and over-capacity warnings.

- [ ] Section 10: private room without owner refused; owner leaving refused; waiting bookings follow a new owner.

12.2 Open questions

1. **Who can send a booking?** This spec assumes every active user. Should it be only users with the Staff role (and the other roles)?
2. **Who prepares the food?** This spec gives Pitika a daily food list (D-22). Is that right, or does someone else need it?
3. **A room, floor, building or office is made inactive** while it has future bookings: cancel them (with a reason shown to the bookers), or refuse until they are moved?
4. **ERD additions:** please confirm the fields marked (added) in section 6: bookings.route, bookings.cancelled_by_id, booking_sessions.requested_start_time and requested_end_time, and NULL allowed in booking_rejections.actor_user_id.
5. **Review history:** is it enough to delete old review rows on a big edit (D-20), or must the history be kept?